Himawari-8 High Temporal Resolution AOD Products Recovery: Nested Bayesian Maximum Entropy Fusion Blending GEO With SSO Satellite Observations

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***Abstract*— High temporal resolution aerosol optical depth (AOD) observations derived from new-generation geostationary (GEO) satellites possess unique advantages in analyzing aerosol fast variation processes and thereby providing more accurate assessments of their climate effects and health risks. Unfortunately, the expected advantages and values are dramatically limited by a relatively large proportion of data missing in the GEO AOD products due to cloud obscuration and intrinsic retrieval algorithm. Although several data recovery algorithms have been proposed in recent years to improve the spatial coverage for GEO AOD products, most of them aim at filling up the data blanks rather than reconstructing the temporally continuous variation of aerosol. Accordingly, in this study, a novel framework of nested spatiotemporal fusion blending GEO with the sun-synchronous orbit (SSO) satellite observations based on the Bayesian maximum entropy (BME) theorem is developed for GEO Himawari-8 Advanced Himawari Imager (AHI) AOD recovery with the sufficient excavation of complementary information from GEO and SSO satellite observations, where the minute-stage and hour-stage BME fusion are jointly employed to reconcile temporal inconsistency and data discrepancies between GEO and SSO observations. The results demonstrate that the AOD spatial coverage is dramatically increased by 240.9% (from 20.5% to 70%) with ensured accuracy after Nested-BME fusion. Additionally, two case analyses, during the development and dispersion processes of haze respectively, both demonstrate that the proposed Nested-BME fusion framework could reconstruct the reliable aerosol diurnal variation trends on the basis of recovering missing data for Himawari-8 AHI AOD datasets, while the AHI official level-2 and level-3 AOD products fail to capture these key trends. Furthermore, the developed Nested-BME AOD fusion framework is also applicable for other GEO satellites over other regions, which could substantially enhance the availability and value of high temporal resolution AOD products for better scientific applications.**

***Index Terms*— Aerosol diurnal variation recovery, aerosol optical depth (AOD) recovery, geostationary (GEO) satellite, nested fusion framework.**

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I. Introduction

ATMOSPHERIC aerosols are solid or liquid particles suspended in the atmosphere that contribute to the global energy budget by directly absorbing and scattering solar radiation and indirectly affecting the lifetime and radiative properties of clouds [1], [2], [3], [4], [5], [6], [7]. In addition, aerosol particles have a significant impact on the ambient atmospheric environment, as aerosol particles dramatically reduce the visibility of the atmosphere and fine aerosol particles cause severe health problems [8], [9], [10], [11]. Generally, aerosol optical depth (AOD),an indicator to quantitatively measure the vertical integral of aerosol extinction within an atmospheric column from the top of the atmosphere to the land surface, is widely applied in related research of atmosphere and environment [12], [13]. Compared with site-specific ground-based aerosol observation, the satellite-retrieved AOD provides an effective and efficient way to measure the spatially continuous distribution of aerosol at a

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large scale. With the launch of new-generation geostationary (GEO) satellites, much finer temporal variation characteristics of the aerosol diurnal cycle can be obtained, contributing significantly to the analysis of aerosol lifecycle and development processes, and thereby providing more accurate assessments of their environmental health risks and climate feedbacks on a regional or global scale [14], [15], [16], [17]. However, the current operational AOD products derived from GEO satellite sensors, such as AHI onboard Himawari-8, generally suffer from substantial data missing, posing a restriction on subsequent research and application [18], [19], [20], [21]. For instance, the majority of AOD data missing is caused by the incapability of passive sensors to obtain AOD values under cloud obscuration conditions [22]. Meanwhile, the intrinsic drawback existing in the aerosol retrieval algorithm is another key factor leading to data missing of AOD products, including but not limited to sun glint, large solar/sensor zenith angle, and ice/snow land surface [23], [24], [25]. Like most of the passive satellite sensors, it is manifested that the spatial coverage of Himawari-8 AOD is poor [13], [26], indicating that the temporally continuous monitoring of diurnal aerosol variation is not available for Himawari-8 on most days of the year, which dramatically restricts its capability and advantage in capturing temporally continuous characteristics of aerosol via high-frequency observations. Therefore, it is meaningful and necessary to conduct data recovery on Himawari-8 AOD products, to enhance its data availability and further restore the spatiotemporal continuity.

In recent years, various data recovery algorithms have been developed to restore missing AOD data. For example, statistic-based and machine learning methods were adopted to predict missing AOD values based on the relationship between AOD value and other factors, such as land use parameters and meteorological factors [27], [28]. However, the universality of the relationship models mentioned above is limited, and the reliability of the end-to-end model is usually affected by the accuracy uncertainty due to over-fitting or under-fitting [12]. Moreover, a variety of approaches have been developed to merge AOD products from different algorithms or different sensors, based on statistical or geo-statistical models. The general statistical methods consist of the polynomial functions, the optimum interpolation, the weighted average model, the least-squares estimation (LSE), the maximum likelihood estimation (MLE), and so forth [29], [30], [31], [32], [33], [34]; while the geo-statistical methods introduce the spatial and temporal autocorrelation into a statistical model, such as universal kriging, spatiotemporal interpolation, Bayesian maximum entropy (BME), and their improved ramification [35], [36], [37], [38], [39], [40], [41], [42]. To our knowledge, the methods mentioned above have been primarily applied to AOD products derived from sun-synchronous orbit (SSO) satellite sensors [e.g., moderate resolution imaging spectroradiometer (MODIS), multiangle imaging spectroradiometer (MISR), visible infrared imaging radiometer suite (VIIRS), and SeaWiFS], for obtaining high spatial coverage AOD datasets on daily basis. As for the AOD recovery on GEO satellites, Li et al. [13] introduced the AOD reanalysis datasets from model simulation [e.g., Modern-Era Retrospective analysis for Research and Applications, version 2 (MERRA-2)] into gap recovery in order to generate full coverage AOD maps. Nevertheless, the accuracy uncertainty from the model-simulated AOD cannot be ignored, and will probably be brought into AOD maps along with the recovery process. Kikuchi et al. [43] fused AHI 10-min AOD data via spatiotemporal interpolation while Lim et al. [44] integrated AOD products from two GEO satellite sensors (i.e., AHI and GOCI), to generate hourly merged AOD dataset, which yielded better accuracy but a limited improvement on spatiotemporal coverage.

Theoretically speaking, the sufficient excavation of complementary information from different categories of satellite sensors could contribute significantly to the improvement of AOD coverage. For example, at any observation moment for a GEO satellite sensor, there generally exists large-area AOD data blank in the sun glint region, which hardly occurs for SSO satellite sensors at the same time. Moreover, it has been manifested that active sensor onboard SSO satellite (e.g., Cloud–Aerosol Lidar with Orthogonal Polarization, CALIOP) can provide strong data supplement for passive satellite sensors by restoring the aerosol information from severe haze or under thin cloud [22], [45]. Although it possesses huge potential to improve the spatial coverage of GEO AOD products by bringing data from SSO satellite sensors in, it faces the challenge of how reconciling temporal inconsistency between GEO and SSO observations. Furthermore, how to establish spatiotemporal correlations through embedding daily SSO data into hourly (even minutely) GEO observations remains to be addressed before recovering the spatiotemporally continuous GEO AOD maps. Besides, data accuracy discrepancies are ubiquitous among different satellite AOD products due to obvious differences in retrieval algorithms and sensor designs, which is nonignorable in the fusion process for ensuring the consistency and reliability of the accuracy of reconstructed AOD values. Therefore, it is required to develop a brand-new framework to address the issues of how to accurately and maximally recover the GEO missing AOD data utilizing complementary information from SSO satellite observations.

The purpose of this study focuses on developing a novel framework of nested spatiotemporal fusion blending GEO with SSO satellite observations to recover the missing pixels in Himawari-8 AHI AOD products as well as reconstruct aerosol diurnal variation trends as much as possible. To be specific, the proposed nested framework is pixel-based AOD recovery on the slide windows of Himawari-8 guided by BME theorem, where the minute-stage and hour-stage spatiotemporal fusion are jointly employed with the sufficient excavation of complementary spatiotemporal information from multisource satellite AOD datasets derived from GEO passive sensor AHI, SSO passive sensor MODIS, and SSO active sensor CALIOP. Besides, the data accuracy inconsistency from multisource satellite AOD products, which resulted from different retrieval algorithms or instrumental designs, is evaluated and incorporated in the fusion process by constructing soft data of original AOD products. The remainder of this article is organized as follows. Section II introduces the study areas and datasets adopted in this study. Section III describes the methodology including data preprocessing and the Nested-BME fusion

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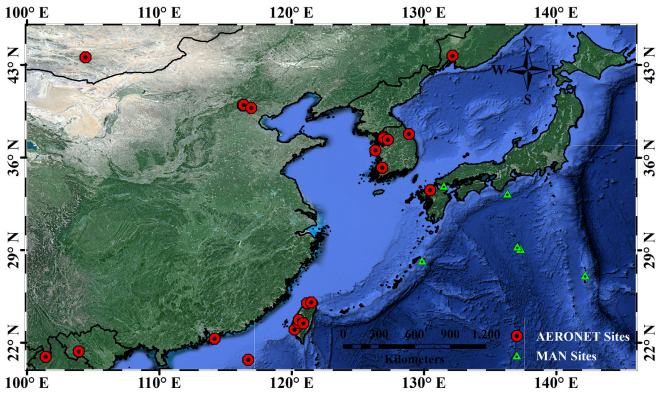


Fig. 1. Experimental area with AERONET site location (illustrated as red octagon) and MAN sites location (illustrated as green triangle) adopted in this study.

framework. Section IV presents the experiment results and discussion, which contains the assessment of accuracy and coverage of AOD recovery, and efficacy evaluation of AOD diurnal variation trend recovery. Finally, the article is concluded in Section V.

II. Study Regions and Datasets

A. Study Regions

In this study, the experimental area mainly comprises East Asia, as shown in Fig. 1, with longitudes ranging from 100°E to 146°E and latitudes ranging from 20°N to 46°N. This area is typically characterized by a high population density and severe air quality issues, which requires accurate and spatiotemporally continuous measurements of aerosols [46], [47], [48]. Meanwhile, there exist diverse and complex elevation variations in this region, typically including the Qinghai–Tibet Plateau and Mongolian Plateau, which could assist to evaluate the universality of the Nested-BME framework.

B. Experimental Datasets

The primary GEO satellite datasets utilized in this article are Himawari-8 AHI aerosol and cloud property (CLP) products with 5-km spatial resolution and 10-min temporal resolution. Meanwhile, the datasets derived from MODIS and CALIOP are selected to typically represent SSO satellite sensors with passive and active observation modes, respectively. In addition, the AOD products from ground-based AEROsol Robotic NETwork (AERONET) and ship-borne Maritime Aerosol Network (MAN) are adopted to validate and evaluate the accuracy and reliability of recovered AOD products. All the datasets mentioned above are listed in Table I and described as follows.

1) Himawari-8 AHI Datasets: Himawari-8, a next-generation GEO meteorological satellite equipped with the AHI sensor, was launched by the Japan Meteorological Agency (JMA) in October 2014 and began operating in July 2015. This instrument was designed with the objective of providing full disk coverage of the Asian-Pacific region with a temporal resolution of 10-min and centered over the equator at approximately 140.7°E [49]. In this study, the AHI-derived AOD data at 500 nm, with 10-min temporal resolution and 5-km spatial resolution, was obtained from the Himawari-8 level-2 latest version 3.0 aerosol property (ARP) products [50], [51]. For quality control, the AOD pixels with a relatively low-quality assurance (QA) flag (QA-flag = "marginal" or "no confidence/retrieval") were excluded during the experiments, and the corresponding 10-min cloud identification extracted from Himawari-8 level-2 version 1.0 CLP products was used to conduct further cloud screening [52]. Besides, the AHI hourly AOD dataset (AOT_merged) from Himawari-8 level-3 version 3.1 ARP product, which was obtained through strict cloud screening and spatiotemporal interpolation on the basis of AHI level-2 ARP product, was adopted as reference and comparison against AOD recovery performance of this study.

2) MODIS Datasets: The MODIS sensors onboard Terra and Aqua satellites with SSO, which crossover the equator around 10:30 and 13:30 local time (L.T.), respectively, provide reliable global coverage observation of aerosol twice a day. The MODIS AOD data retrieved via the dark target (DT) algorithm generally performs well over densely vegetated regions, while the deep blue (DB) algorithm-derived AOD data has unique advantages to derive aerosol information over bright land surfaces except for snow/ice cover [53], [54], [55]. In this study, the MODIS DT–DB combined AOD at 550 nm with 10 km spatial resolution and relatively high QA flag, which was extracted from the MODIS Collection 6.1 level-2 aerosol products (MOD04_L2 for Terra and MYD04_L2 for Aqua), was adopted to obtain AOD datasets with reliable accuracy and larger spatial coverage [56], [57], [58]. Furthermore, the MODIS 550 nm AOD was converted to the same wavelength of AHI AOD as 500 nm, by calculating the corresponding Ångström exponent through logarithmic interpolation from AOD retrievals at 470 and 550 nm.

3) CALIOP Datasets: The CALIOP onboard the Cloud–Aerosol Lidar and Infrared Pathfinder Satellite Observation (CALIPSO) is a satellite sensor with active observation mode launched in 2007 as a member of the A-train constellation, which could provide atmospheric profile information at approximately 1:30 and 13:30 L.T. In this study, the daytime and nighttime CALIOP AOD data are obtained from the CALIOP level-2 latest version 4.2 aerosol profile products (APro), which could provide more reliable AOD retrievals than those of previous versions due to the improved algorithm such as the reconsideration of elevated aerosols and lidar ratio [59], [60], [61], [62]. For quality control, the AOD retrievals were excluded by evaluating the corresponding atmospheric volume description (AVD) when the CALIOP profile was unable to reach the land surface due to the cloud or dense aerosol layer, specific incapability to capture the signal from the land surface, or near-surface aerosol scattering [63], [64], [65]. Moreover, the daytime cloud/aerosol identification data was obtained from CALIOP level-2 version 4.2 vertical feature mask products (VFM), which describe morphological and vertical characteristics of detectable atmospheric features, in order to provide more precise cloud screening (e.g., thin cloud or cloud edge circumstances) for corresponding daytime AOD datasets from the passive sensors.

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TABLE I
SUMMARY OF THE EXPERIMENTAL DATASETS ADOPTED IN THE NESTED-BME FRAMEWORK

Instrument	Dataset	Spatial Resolution	Temporal Resolution	Description
Himawari-8 AHI	ARP_L2	5 km	10 min	Version 3.0
	CLP_L2	5 km	10 min	Version 1.0
Terra MODIS	MOD04_L2	10 km	daily (a.m.)	Collection 6.1
Aqua MODIS	MYD04_L2	10 km	daily (p.m.)	Collection 6.1
CALIOP	CAL_LID_L2_05kmAPro	5 km	daytime/nighttime	Version 4.2
	CAL_LID_L2_VFM	5 km	daily (p.m.)	Version 4.2
AERONET	Level 2.0	-	-	Version 3.0
MAN	Level 2.0	-	-	Version 2.0

4) AERONET/MAN AOD Datasets: AERONET is a ground-based aerosol observation network, generally providing AOD measurements via sun-photometer CE-318, which is widely acknowledged as "ground truth" to evaluate satellite-derived AOD products [66], [67], [68], [69]. MAN is a ship-borne aerosol observation network, offering periodical AOD observations via handheld sun photometers Microtops-II over sea surfaces. This sea-level-based measurement served as a supplementary component to the ground-based observations in the whole AERONET [70]. In this study, the level 2.0 AERONET AOD products from 22 AERONET sites and MAN aerosol observations in East Asia were adopted during the experimental period, which is illustrated in Fig. 1 and listed in supporting information (SI) Tables S1 and S2 with detailed elevation and geographic locations. The level 2.0 AERONET AOD products are cloud-screened and quality assured with an AOD uncertainty of less than 0.02, which is used for evaluation and validation of original satellite-derived AOD products and recovered AOD data in this study. Since a part of AERONET sites usually provides AOD measurements at multichannels except for 500 nm, 500 nm AERONET AOD was interpolated using AERONET AOD at adjacent channels based on second-order polynomial fitting [71], [72].

III. METHODOLOGY

The Nested-BME fusion framework developed in this study is pixel-based AOD recovery on the slide spatiotemporal window of Himawari-8, blending AHI with MODIS/CALIOP satellite observations based on BME theorem, aiming at restoring hourly AOD datasets and reconstructing the diurnal variation trends of aerosol as much as possible. As depicted in the integrated flowchart in Fig. 2, the Nested-BME fusion framework primarily contains three-stage procedures, namely data preprocessing, minute-stage BME fusion, and hour-stage BME fusion. The data preprocessing is first performed to yield spatiotemporal consistency for multisource satellite products, as well as to ensure their data availability and reliability of accuracy. Then, the minute-stage BME fusion is conducted to obtain hourly AOD datasets with a certain recovery effect by fully utilizing the temporal correlation and data availability derived from AHI 10-min AOD products. If the temporal information from minute-stage BME fusion is insufficient to conduct reliable AOD recovery, the hour-stage BME spatiotemporal fusion is implemented to accurately and maximally recover the missing AOD data, primarily based on complementary information from SSO satellite observations and reconstruction of spatiotemporal correlations. The minute-stage and hour-stage BME fusion are jointly employed in our framework in order to ensure the missing data in AHI hourly AOD is recovered as much as possible, as well as the spatiotemporally continuous characteristics of aerosol. The specific procedures of the Nested-BME fusion framework are described as follows.

A. Data Preprocessing

To ensure both spatial and temporal consistency of different data categories, as well as their reliability of accuracy, a sequence of preprocessing procedures has been conducted on Himawairi-8 AHI original AOD products, MODIS AOD data, and CALIOP datasets. In terms of spatial unification, the auxiliary SSO datasets, including the MODIS-derived products and CALIOP-derived data, have been first reprojected into the World Geodetic System 1984 geographic coordinate system and then resampled using bilinear interpolation to match the AHI geographic grids with 5 km spatial resolution. From the perspective of temporal integration, both the MODIS and CALIOP AOD datasets are allocated and spliced to obtain hourly AOD datasets within 30 min before and after corresponding to that hour, elaborately matching the temporal resolution for the hour-stage BME fusion framework. Moreover, the morphological dilation to 1° centered on the profiling track has been conducted on CALIOP AOD and cloud identification datasets due to limited spatial coverage of CALIOP observation, which was described in detail in our previous research [22]. Furthermore, it is crucial to reduce the impacts from error AOD pixels caused by cloud misidentification as aerosol as much as possible, since avoiding probably propagation of abnormal values from original AOD products is essential during the recovery processes. Therefore, the original AHI 10-min AOD products ($AOD_{original}$) have been first screened by the corresponding AHI cloud mask, then strict cloud screening has been conducted on each AHI 10-min AOD by excluding the pixels that exceed the

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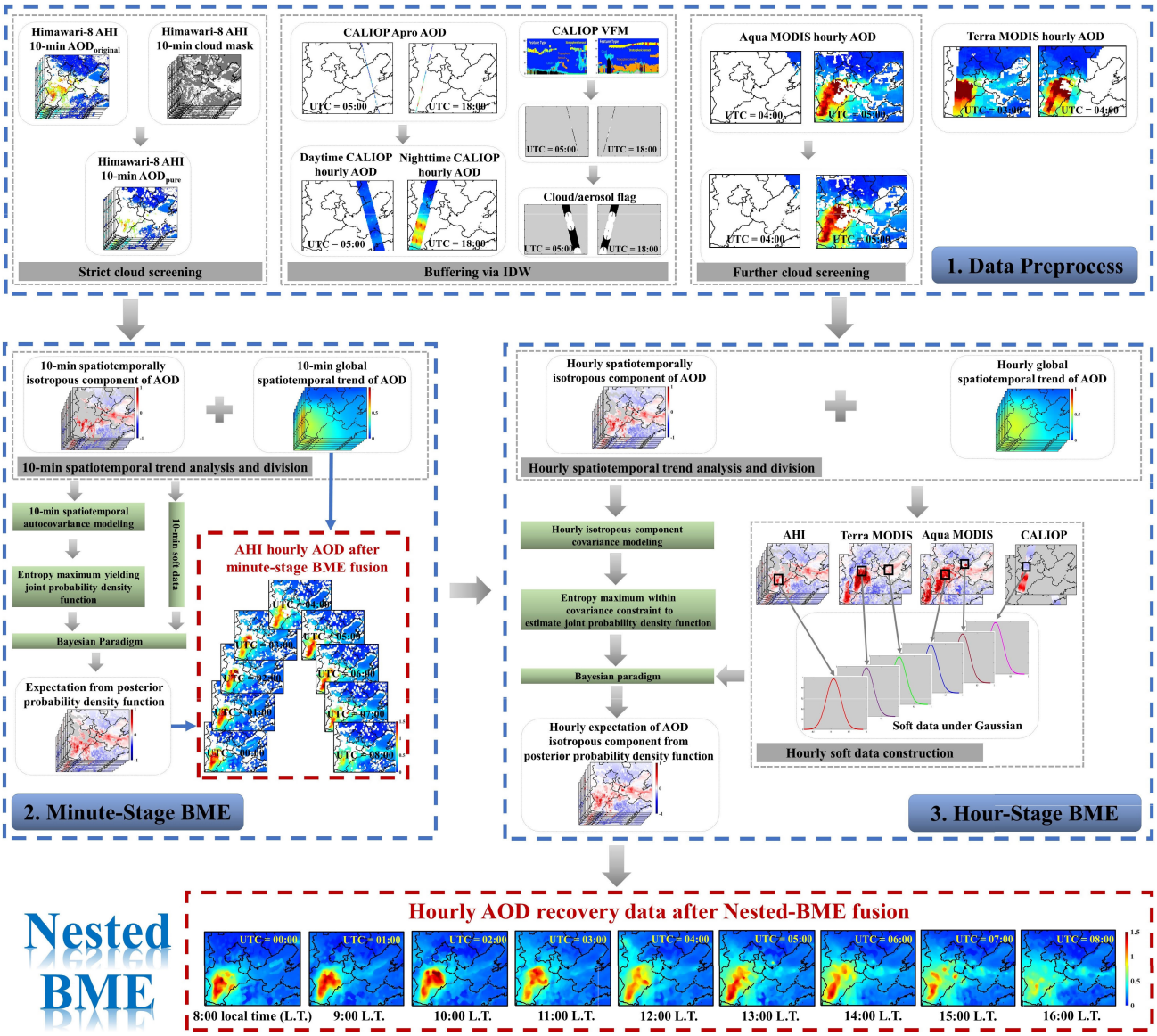


Fig. 2. Flowchart of the proposed Nested-BME fusion framework blending GEO with SSO satellite observations to obtain hourly AOD recovery datasets for GEO satellite.

threshold of error variance, which is estimated and calculated by adjacent spatiotemporal AOD pixels as part of Himawari-8 official level-3 AOD retrieval algorithm [43], in order to obtain AHI 10-min AOD datasets (AOD_{pure}) excluding cloud contamination as far as possible. Meanwhile, the daytime cloud identification data from CALIOP has been utilized to conduct supplementary cloud screening on the AHI hourly AOD dataset after minute-stage recovery and Aqua MODIS AOD product, due to the advantage of CALIOP over passive satellite sensors in identifying aerosol and cloud.

B. BME Principle

The BME is a universal and versatile framework, which provides a method to integrate and further analyze the spatiotemporal data with various knowledge bases and different extents of epistemic uncertainty [73], [74]. In this study, the expectation value of a missing AOD pixel $\bar{x}_{AOD}$ on the grid (x, y) at time t could be nonlinearly estimated from several spatiotemporally adjacent satellites-derived AOD observations with irregular spatial or temporal gaps, based on the following equation:

$$\bar{x}_{AOD} = \int x_{AOD} f(x_{AOD} | x_{Soft_AOD,1}, x_{Soft_AOD,2}, \ldots, x_{Soft_AOD,n}) dx_{AOD} \quad (1)$$

where x_{Soft_AOD} represent the probabilistic AOD soft data derived from spatiotemporally adjacent valid satellite AOD datasets, while $f(x_{AOD} | x_{Soft_AOD,1}, x_{Soft_AOD,2}, \ldots, x_{Soft_AOD,n})$ represent the posterior probability density function for the center pixel x_{AOD} under the spatiotemporally adjacent pixel observations, which is estimated by updating the prior probability density function along with the introduction of soft data. The posterior probability density function can be expressed as

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(2) following the Bayesian rule [75]:

$$f\left(x_{\mathrm{AOD}} | x_{\mathrm{Soft_AOD},1}, x_{\mathrm{Soft_AOD},2}, \ldots, x_{\mathrm{Soft_AOD},n}\right) = \frac{f_G\left(x_{\mathrm{map}}\right)}{f\left(x_{\mathrm{Soft_AOD},1}, x_{\mathrm{Soft_AOD},2}, \ldots, x_{\mathrm{Soft_AOD},n}\right)}. \quad (2)$$

The term $f\left(x_{\mathrm{Soft_AOD},1}, x_{\mathrm{Soft_AOD},2}, \ldots, x_{\mathrm{Soft_AOD},n}\right)$ represents the prior probability density function of the spatiotemporally adjacent AOD soft data, the x_{map} is an aggregation of all the valid spatiotemporally adjacent AOD soft data, and the $f_G(x_{\mathrm{map}})$ is the joint probability density function and estimated by achieving the entropy maximum under the constraint of one particular general knowledge [76], [77], [78], which is expressed as G in the (2). The general knowledge in this study, namely term $G\left(x_{\mathrm{map}}\right)$, is defined as a statistical spatiotemporal covariance model for autocorrelation of AOD soft data. And the joint probability density function can be solved by bringing in a Lagrange multiplier λ to maximize entropy yielding the lowest prediction risk [78], [79], which is expressed in the following:

$$f_G\left(x_{\mathrm{map}}\right) = \frac{\exp\left[\sum_{\alpha=1}^{n} \lambda_\alpha G_\alpha\left(x_{\mathrm{map}}\right)\right]}{\int \exp\left[\sum_{\alpha=1}^{n} \lambda_\alpha \mathrm{G}_\alpha\left(x_{\mathrm{map}}\right)\right] \mathrm{d}x_{\mathrm{map}}}. \quad (3)$$

Finally, the joint probability density function $f_G\left(x_{\mathrm{map}}\right)$ can be substituted in (1) and (2) to estimate the expectation value of a missing AOD pixel $\bar{x}_{\mathrm{AOD}}$.

C. Trend Analysis and Trend Removal

To meet the requirement of BME theory that all the variables are regionalized random to construct a spatiotemporal random field model with homogeneous and isotropous field [80], [81], [82], the analysis and removal of a global spatiotemporal trend are essential in order to obtain spatiotemporal autocorrelation characteristics of satellite AOD products [75]. In this study, the global spatiotemporal trend of AOD datasets is quantified by a slide spatiotemporal window filter centered on each AOD pixel. For each pixel at the grid (x, y) on observation time T, its global spatiotemporal trend component $\mathrm{AOD}_{\mathrm{trend}}$ is estimated by averaging available AOD pixels within the spatiotemporal window with a size of 99 × 99 × 3 as follows:

$$\mathrm{AOD}_{\mathrm{trend}} = \frac{1}{N}\left(\sum_{i=-49}^{i=49}\sum_{j=-49}^{j=49}\sum_{t=-1}^{t=1} \mathrm{AOD}_{x+i,y+j,T+t}\right) \quad (4)$$

where N is the number of AOD pixels with valid values within the spatiotemporal window. The range of T includes the previous and next 10 min of observation time in the minute-stage BME fusion and represents the temporal window containing the previous and next 1 h of observation time during the hour-stage BME fusion. This spatiotemporal window filter keeps moving until the global spatiotemporal trend component for each AOD pixel is calculated.

Then, the spatiotemporally isotropous component of AOD could be obtained by subtracting the global spatiotemporal trend from the original AOD datasets. This spatiotemporally isotropous component of AOD is expected to be spatiotemporally stationary with an approximately normal distribution, which fulfills the requirements for the following spatiotemporal covariance model construction and expectation estimation.

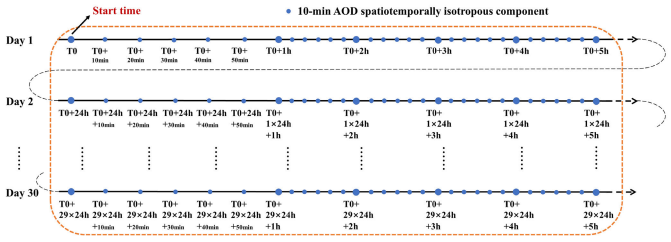


Fig. 3. Schematic of minute-stage spatiotemporal covariance model construction in one month.

D. Soft Data Construction

The accuracy inconsistency resulting from different retrieval algorithms or instrumental designs is ubiquitous among different satellite AOD products, which is nonignorable in the multisource fusion process for ensuring the consistency and reliability of the accuracy of reconstructed AOD values. In this study, the Gaussian distribution probability soft data is generated for original multisource satellite AOD products, and is constructed through comparison against AERONET AOD products [42]. For an AOD product derived from any type of satellite sensor, its general bias ε could be calculated and expressed as

$$\mathrm{AOD}_{\mathrm{sat},i} = \mathrm{AOD}_{\mathrm{AERONET},i} + \varepsilon_i, \quad \varepsilon \sim N\left(\mu_\varepsilon, \sigma_\varepsilon^2\right) \quad (5)$$

where $\mathrm{AOD}_{\mathrm{sat},i}$ and $\mathrm{AOD}_{\mathrm{AERONET},i}$ represent one pair (i) of satellite AOD pixel value and corresponding AERONET AOD observation, while the ensemble of these errors ε generally possesses the characteristic of independency and normally random distribution. Thus, the Gaussian distribution probability soft data for the satellite AOD after global spatiotemporal trend removal could be constructed using the following equation:

$$\mathrm{AOD}_{\mathrm{soft,sat}} \sim N\left(\mathrm{AOD}_{\mathrm{sat}} + \mu_\varepsilon, \sigma_\varepsilon^2\right) \quad (6)$$

where $\mathrm{AOD}_{\mathrm{soft,sat}}$ is the Gaussian distribution probability soft data for spatiotemporally isotropous component of AHI, MODIS, or CALIOP AOD. The term $\mathrm{AOD}_{\mathrm{sat}}$ is the hourly spatiotemporally isotropous component of AHI, MODIS, or CALIOP AOD, while the terms μ_ε and σ_ε^2 present the mean and variance of Gaussian distribution probability soft data, respectively.

E. Spatiotemporal Covariance Modeling

The general knowledge $G\left(x_{\mathrm{map}}\right)$ in this study, which is the ensemble of AOD data with specific spatial coordinates and observation time possessing the characteristics of spatiotemporal variation, could be expressed as a spatiotemporal covariance function described by the distance of space and time [83], [84]. To be specific, the spatiotemporal covariance function describes the spatiotemporal dependency of existing AOD data using statistical models and then estimates the expectation of a random field for missing AOD pixels.

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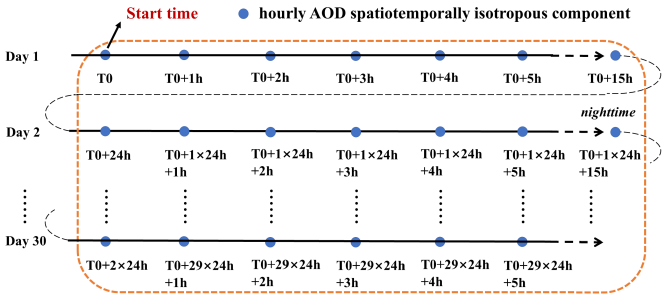


Fig. 4. Schematic of hour-stage spatiotemporal covariance model construction in one month.

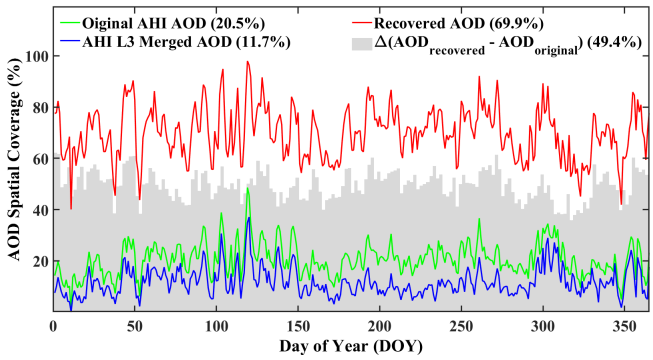


Fig. 5. Time series plot of daily averaged AOD coverage for original AHI L2 AOD product, recovered AOD after Nested-BME fusion, and AHI L3 merged AOD product.

In this study, the ensemble of 10-min AOD datasets for minute-stage modeling is conducted by constructing continuous temporal dimension for all the 10-min AOD observations in one month, which is illustrated in Fig. 3 on the assumption that this month has 30 days with 5 h valid AOD observations every day. While the ensemble of hour AOD datasets for hour-stage modeling is conducted by constructing continuous temporal dimension for multisource AOD observations in one month, which is illustrated in Fig. 4 on the assumption that this month has 30 days with 5 h valid AOD observations at daytime and 1 h valid AOD observations at nighttime. Moreover, a nested spatiotemporal covariance model, which is expressed as (7) including two exponential models proportionally summed up, is adopted to describe the complex spatiotemporal variation mechanism [85], [86].

$$G_{\tau=|t-t'|}^{d=|s-s'|} = c_1 \exp\left(-\frac{3d}{m_{s1}}\right)\exp\left(-\frac{3\tau}{n_{t1}}\right) + c_2 \exp\left(-\frac{3d}{m_{s2}}\right)\exp\left(-\frac{3\tau}{n_{t2}}\right). \quad (7)$$

In the equation above, d and τ are the distance of space and time between AOD pixel pairs on spatiotemporal coordinates (s, t) and (s', t'), respectively. c_1 and c_2 are the partial sill variances of the two separate exponential models. m_{s1} and m_{s2} are the spatial correlation ranges for the two exponential models, while n_{t1} and n_{t2} are the temporal correlation ranges for these two exponential models, respectively. The parameter fitting of covariance for 10-min and hourly AOD spatiotemporally isotropous components are conducted separately for each month, and all the monthly model parameters for minute-stage and hour-stage BME in this study are listed in SI Tables S3 and S4, respectively.

F. Recovered AOD Estimation

After constructing the spatiotemporal covariance model for AOD spatiotemporally isotropous component, a Lagrange multiplier λ is introduced to maximize entropy for estimating the joint probability density function as (3). Then the hourly expectation from the posterior probability density function is calculated by coupling the AOD soft data into the joint probability density function based on the Bayesian paradigm as (2). The spatiotemporal neighborhood searching ranges are chosen as 5 km and 60 min, respectively, during the integration of 10-min AOD soft data, which corresponds to one pixel surrounding the target pixel within 1 h. The spatiotemporal searching radius in hour-stage BME fusion is determined by spatial and temporal correlation ranges in spatiotemporal covariance model fitting. Finally, the recovered AOD data is obtained by adding the hourly expectation from the posterior probability density function to the global spatiotemporal trend at the corresponding pixel and time.

IV. Experiment Results and Discussion

A. Assessment on the Performance and Coverage of AOD Recovery

The comparisons of daily averaged coverage are conducted between the original AHI L2 AOD product, recovered AOD data after Nested-BME fusion, and AHI L3 merged AOD product, which is illustrated in Fig. 5 marked as green line, red line, and blue line, respectively. The daily averaged coverage is calculated by averaging the valid AOD completeness percentage of the experimental region for each daytime hour on that day. As shown in Fig. 5, the annually averaged spatial coverage of the original AHI L2 AOD product is only about 20.5%. In contrast, the annually averaged spatial coverage of recovered AOD data after Nested-BME fusion reaches nearly 70%, which dramatically increased by almost 50% absolute coverage compared with the original AHI L2 AOD product. Meanwhile, the absolute improvement of daily averaged spatial coverage, represented as the gray areas in Fig. 5, ranges roughly from 40% to 60% with an average of 49.4%. Moreover, the daily averaged spatial coverage of the AHI L3 merged AOD product is shown as a reference, with annually averaged spatial completeness of only 11.7%, even less than those of the original AHI L2 AOD, which is a primary reason for choosing not to adopt this dataset directly in the fusion.

Furthermore, a comparison is conducted between the original AHI L2 AOD and the recovered AOD after Nested-BME fusion through annual coverage maps. As illustrated in Fig. 6(a), the annual averaged coverage ratio of original AHI L2 AOD products ranges from 0% to 44.4% with a median of 20.2%, and the lowest coverage ratio of original AOD primarily is found in the Southwestern regions of China and Northern areas of Japan, followed by northern China

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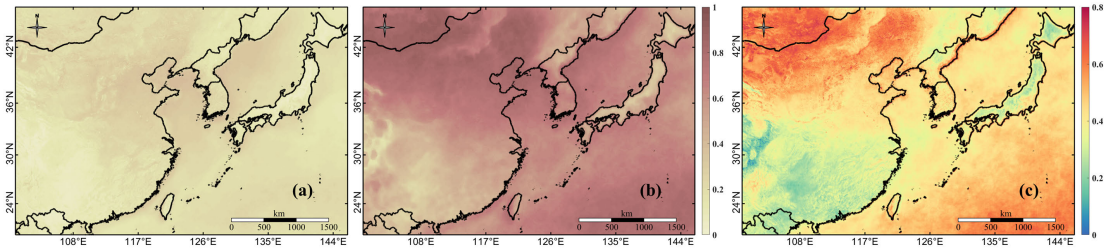


Fig. 6. Maps of annual AOD coverage for (a) original AHI L2 AOD product and (b) recovered AOD after Nested-BME fusion, with (c) coverage improvement between them. The color bars indicate the completeness for (a) and (b) and completeness improvement for (c).

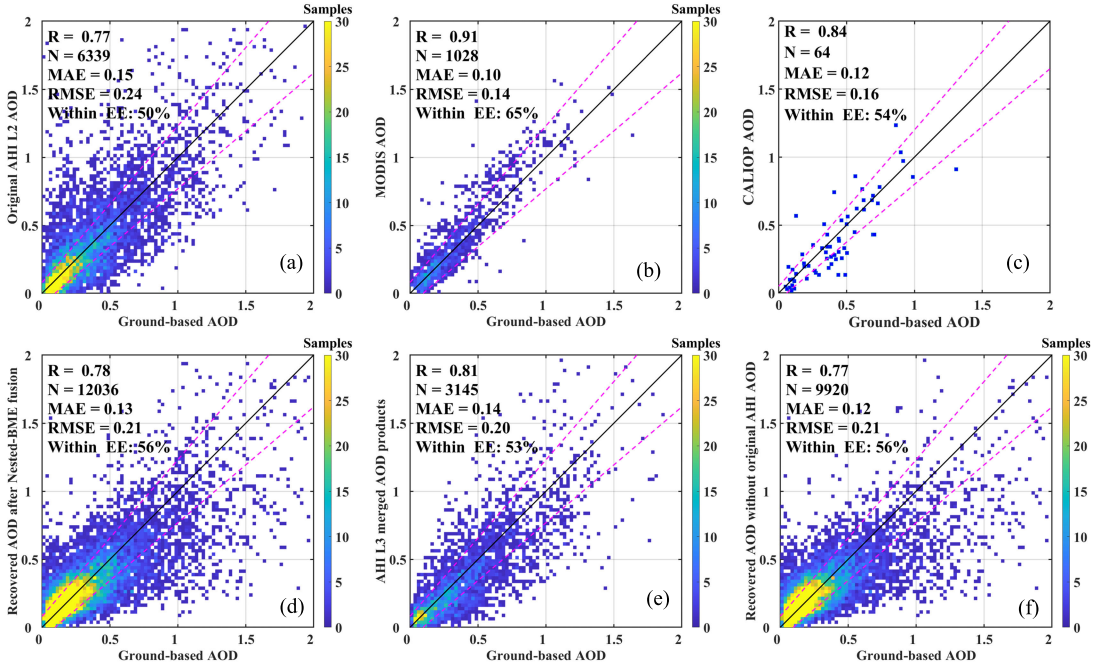


Fig. 7. Accuracy validation results of original satellite AOD products and recovered AOD datasets against AERONET AOD observations within the study area during the experimental period for (a) original AHI L2 AOD product, (b) original MODIS AOD product, (c) original CALIOP AOD product, (d) overall recovered AHI AOD dataset after Nested-BME fusion, (e) officially released AHI L3 merged AOD product, and (f) recovered AHI AOD dataset without original AHI AOD. The black solid line represents the 1:1 reference line, the region within the magenta dashed lines represents the EE envelop, and the colorbar represents the density of samples of AOD matchups.

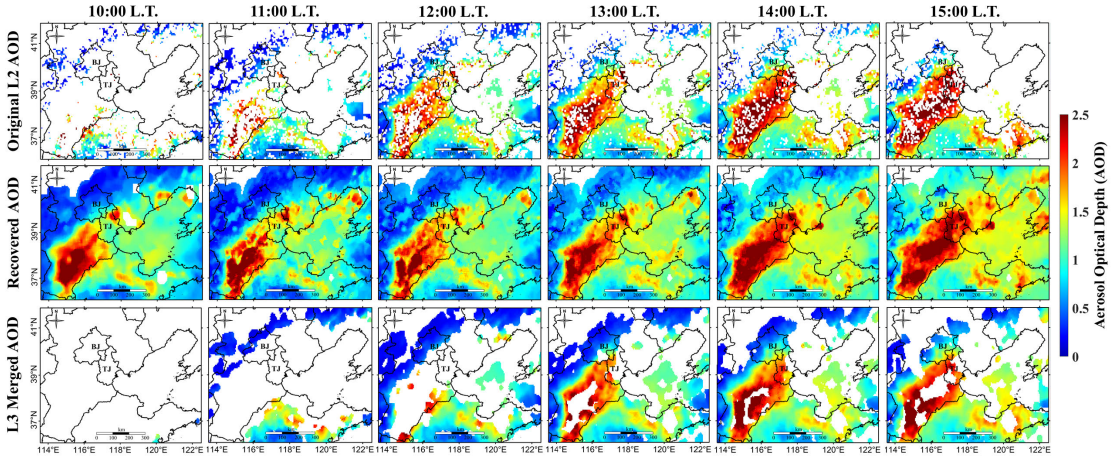


Fig. 8. AOD maps from 10:00 to 15:00 L.T. over BTH regions on January 25, 2017, extracted from original AHI L2 AOD product (first row), recovered AOD dataset after Nested-BME fusion (second row), and officially released AHI L3 merged AOD product (third row), respectively.

and southern Russia within the experimental areas. In contrast, the annual averaged coverage ratio of recovered AOD datasets ranges from 12.5% to 91.37% with a median of 64.6%, which is illustrated in Fig. 6(b) with similar spatial patterns as that of original AOD products, indicating that the data completeness after Nested-BME fusion depends on

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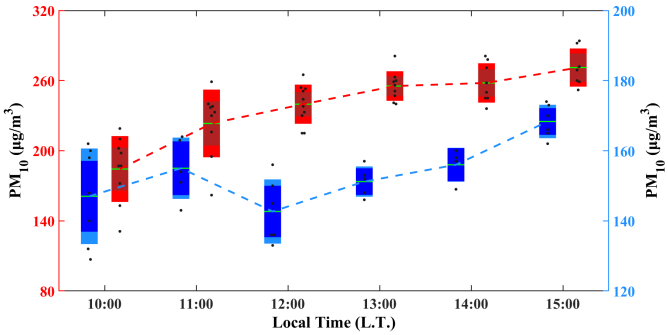


Fig. 9. Diurnal variation of ground-monitor PM_{10} on January 25, 2017 over Beijing (blue) and Tianjin (red), respectively. The black solid dots represent PM_{10} measurements from different ground-monitor sites of that city, and each boxplot contains mean value, and 10th, 25th, 75th, and 90th percentiles of PM_{10} measurements from all monitor sites in that hour.

the spatiotemporal coverage of original AOD products to a certain extent. Besides, as demonstrated in Fig. 6(c), there exists substantial improvement for the AOD completeness in the experimental regions, especially in the South of Mongolia, North of China, and Western Pacific region near the satellite nadir. The probable reason is that the original AHI AOD products periodically have intrinsic data blanks when close to the satellite nadir due to the sun glint effect or far from the satellite nadir due to the combination of large viewing zenith and high surface reflectance [87], which could be well recovered by complementary AOD observations derived from SSO satellites.

B. Evaluation of the Accuracy of AOD Recovery

The AERONET AOD is adopted as a true reference to validate the recovered AOD dataset after Nested-BME fusion and compare its accuracy with that of the original AHI, MODIS, and CALIOP AODs. As the AERONET sites provide ground-level point measurements, the AOD values of both original and merged data are extracted by averaging 3 × 3 grids centered on the grid where the AERONET site is located. In order to acquire a more reasonable comparison, the matchups with an average of AERONET AOD observations within ±30 min are adopted to validate the original satellite AOD products and recovered AOD datasets. The validation results are assessed via the indicators including the mean-absolute-error (MAE), the root-mean-square error (RMSE), and the expected error (EE) envelop, which are defined in the following equations [56], [88], respectively, where the term $AOD_{org/recov}$ represents the original satellite AOD product or merged AOD data, and the term $AOD_{AERONET}$ represents the ground monitoring AOD data from AERONET sites:

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|AOD_{org/recov} - AOD_{AERONET}\right| \tag{8}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(AOD_{org/recov} - AOD_{AERONET}\right)^2} \tag{9}$$

$$EE = \pm 0.05 \pm 0.15 \times AOD_{AERONET}. \tag{10}$$

The accuracy validation results of original AOD products or recovered AOD datasets against AERONET AOD observations within the study area during the experimental period are shown and compared in Fig. 7. Fig. 7(a)–(c) illustrates the accuracy of original AHI L2 AOD products, and original MODIS and CALIOP AOD products, respectively, which serve as references to evaluate the AOD recovery accuracy. As shown in Fig. 7(d), the overall recovered AOD dataset after Nested-BME fusion performs well with the correlation coefficient (R) of 0.78, the MAE of 0.13, the RMSE of 0.21, and approximately 56% of the matchups falling within the EE envelop for a total count of 12 036 matched points, which demonstrate improved accuracy compared to the original AHI AOD dataset, benefited from the strict cloud screening process and the relatively higher data accuracy from MODIS and CALIOP AOD products. Meanwhile, the validation of recovered AODs where the original AHI AOD is missing, which is demonstrated in Fig. 7(f), possesses a similar accuracy as the overall recovered AOD dataset, indicating that the errors introduced during the Nested-BME fusion process are slight and acceptable. In contrast, the validation result of the officially released AHI L3 merged AOD product, which is shown in Fig. 7(e), exhibits a comparable data accuracy as the recovered AOD dataset in this study. However, the number of matchups of the AHI L3 merged AOD product is much less than those of the recovered AOD dataset or even the original AHI L2 AOD product, which is also reflected from the coverage comparison in Fig. 5, suggesting that the officially released AHI L3 merged AOD product improves the data accuracy via multitemporal fusion nevertheless at the expense of data spatiotemporal completeness. Overall, it is indicated that the Nested-BME fusion framework provides a reliable and robust AOD recovery algorithm for GEO satellite Himawari-8, which could dramatically improve the overall completeness of the AHI AOD dataset from 20% to approximately 70% with ensured data accuracy.

C. Evaluating the Recovery Performance of AOD Diurnal Variation Trends

In addition to the improvement of spatial coverage on the premise of guaranteed recovery accuracy, the recovery of the AOD diurnal variation trend is also of vital significance, since one of the principal advantages of GEO satellites over SSO satellites in atmospheric aerosol observations is their ability to capture aerosol diurnal variations through their high-frequency observation capabilities. Thus, in order to further demonstrate the efficacy and reliability of AOD diurnal variations recovery derived from the Nested-BME fusion process, two case analyses have been conducted through a haze development process in Beijing–Tianjin–Hebei (BTH) regions and a haze dispersion process in Yangtze River Delta (YRD) regions, respectively.

Fig. 8 demonstrates a case of the haze development process in BTH regions, which suffered from severe atmospheric pollution on January 25, 2017. As illustrated in the first row of Fig. 8, there exists large-area AOD data missing in the original AHI L2 AOD products over the majority of regions of Beijing and Tianjin in the morning of this case day, subject

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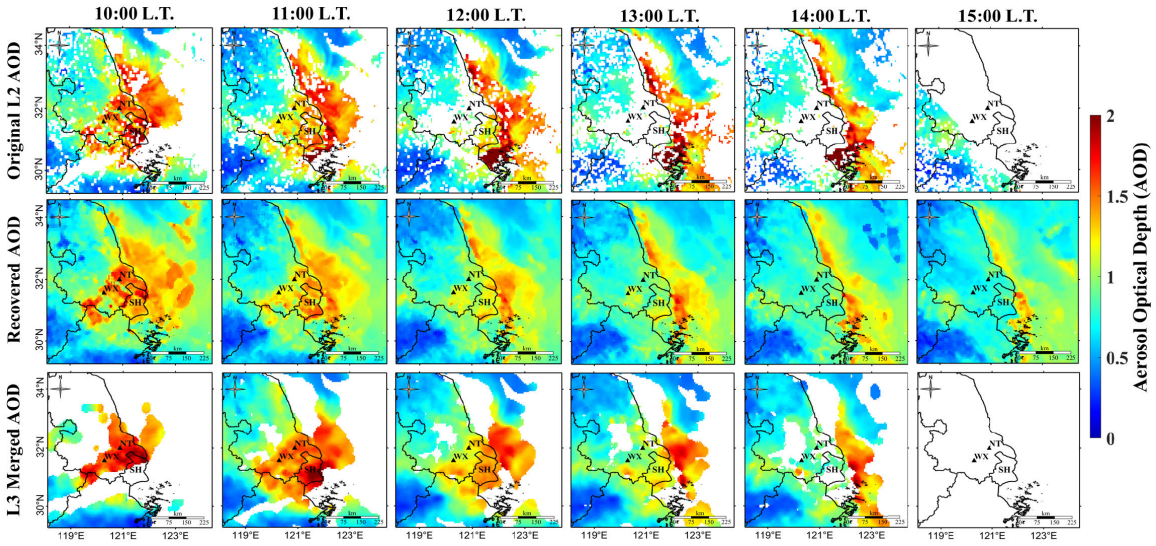


Fig. 10. AOD maps from 10:00 to 15:00 L.T. over YRD regions on December 24, 2017, extracted from original AHI L2 AOD product (first row), recovered AOD dataset after Nested-BME fusion (second row), and officially released AHI L3 merged AOD product (third row), respectively.

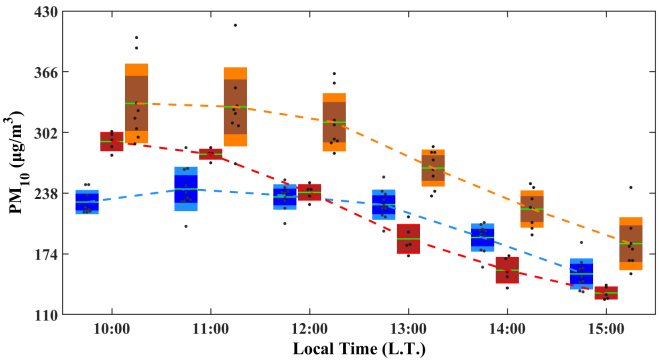


Fig. 11. Diurnal variation of ground-monitor PM_{10} on December 24, 2017 over Shanghai (blue), Nantong (red), and Wuxi (orange), respectively. The black solid dots represent PM_{10} measurements from different ground–monitor sites of that city, and each boxplot contains mean value, and 10th, 25th, 75th, and 90th percentiles of PM_{10} measurements from all monitor sites in that hour.

to the combined factors of cloud obscuration, severe haze, and solar zenith condition. This AOD data missing issue is not essentially resolved through the spatiotemporal merging process adopted in the officially released AHI L3 merged AOD products, which is illustrated in the third row of Fig. 8. According to Fig. 9, which depicts the hourly variation of ground-monitor PM_{10} concentrations over Beijing and Tianjin, both cities exhibited a trend of increasing atmospheric pollution. However, it is difficult to obtain diurnal variation patterns for aerosol over BTH regions, by only using the fragmented AOD maps derived from AHI official L2/L3 AOD products, which subsequently fails to capture the haze development trend. In contrast, after the AOD recovery via the Nested-BME fusion process, most of the missing data has been restored in the recovered AOD maps, which is shown in the second row of Fig. 8. In addition to that, as can be seen from the temporally continuous AOD maps derived from recovered AOD datasets after Nested-BME fusion, southern Hebei region suffered from severe haze events throughout the entire day, while the Beijing and Tianjin experienced gradual increases in the severity extent of atmospheric pollution from morning to afternoon. And the recovered observation-based aerosol temporal variation pattern is consistent with the diurnal variation trend of monitored PM_{10} on this case day, indicating that the proposed AOD recovery approach for GEO satellite Himawari-8 possesses the capability to restore the haze development process with reliability.

As illustrated in Fig. 10, another case analysis of the haze dispersion process is conducted over YRD regions on December 24, 2017. As shown in the first row of Fig. 10, there exists a certain amount of missing data in the AOD maps derived from original AHI L2 AOD products throughout the entire day, which is primarily caused by thin cloud contamination. Meanwhile, the large-area AOD missing situations usually appear early in the morning or late in the afternoon (e.g., 15:00 L.T. in this case), since the solar zenith conditions that satisfy the requirements for aerosol retrieval algorithm are generally limited in the winter of the northern hemisphere. As shown in the third row of Fig. 10, in addition to failing to address the AOD data missing issue, the officially released AHI L3 merged AOD products further screen some AOD data from L2 original AOD products via spatiotemporal merging due to suspicious thin cloud contamination. According to hourly variation of ground-monitor PM_{10} concentrations over YRD regions as shown in Fig. 11, a decreasing trend of atmospheric pollution is found, which is hard to be obtained via AHI official L2/L3 AOD products that subsequently fail to capture the haze dispersion process. In contrast, the AOD maps after the AOD recovery via the Nested-BME fusion process have restored most of the missing data, as shown in the second row of Fig. 10. Moreover, the temporally continuous recovered AOD maps illustrate that the aerosol loadings over YRD regions (especially the Shanghai, Nantong, and Wuxi cities) demonstrate a declining trend from morning to afternoon, which is consistent with diurnal variation trend of

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monitored PM_{10} on this case day, indicating that the proposed AOD recovery approach performs well in restoring the haze dispersion process to a certain extent with reliability. Overall, it is proven that the proposed framework of Nested-BME fusion blending GEO with SSO satellite observations could help to reconstruct aerosol diurnal variation trends on the basis of recovering missing data for Himawari-8 AHI AOD datasets.

V. Conclusion and Future Work

A significant proportion of data missing in the GEO AOD dataset reduces its ability to capture temporally continuous characteristics of aerosol via high-frequency observations, which further limits its value in atmosphere-related research and applications. In this study, we develop a novel framework of nested spatiotemporal fusion blending GEO with SSO satellite observations to recover the missing pixels in Himawari-8 AHI AOD products and reconstruct aerosol diurnal variation trends. To be specific, the proposed nested framework is pixel-based AOD fusion on the slide windows of Himawari-8 guided by BME theorem, where the minute-stage and hour-stage spatiotemporal fusion are jointly employed with the sufficient excavation of spatial and temporal correlation from multisource satellite AOD data. The results demonstrate that the annually averaged spatial coverage of the original AHI L2 AOD product is only approximately 20.5%, while the annually averaged spatial coverage of recovered AOD data after Nested-BME fusion reaches nearly 70%, which is dramatically increased by 240.9% compared with the original AHI L2 AOD product. The extent of coverage improvements varies from region to region, generally depending on the original spatial and temporal coverage, while dramatic coverage improvements can be expected over regions affected by sun glint and large viewing zenith, due to the complementary AOD observations derived from SSO satellites. Moreover, the recovered AOD datasets demonstrate improved accuracy compared to the original AHI AOD dataset, benefited from the data screening on the original AHI AOD products and the relatively higher data accuracy from MODIS and CALIOP AOD products, which indicates that the Nested-BME fusion framework is a reliable AOD recovery algorithm dramatically improving the overall completeness of AHI AOD dataset with ensured data accuracy. Furthermore, two case analyses in BTH and YRD regions demonstrate that the proposed framework of nested spatiotemporal fusion blending GEO with SSO satellite observations could not only improve the spatiotemporal coverage for AHI AOD datasets on the premise of accuracy assurance, but also help to reconstruct aerosol diurnal variation trends, such as restoring the development and dispersion processes of haze.

In this study, we select Himawari-8 AHI and MODIS/CALIOP as representative sensors of the GEO and SSO satellites to demonstrate the advantages and performances of the Nested-BME AOD fusion framework. Obviously, the developed Nested-BME framework is also applicable for other GEO satellites, such as the FY-4A and GOES-R series, to substantially enhance the availability of high temporal resolution AOD products for better scientific applications. Admittedly, the spatiotemporal coverage of recovered AOD datasets in this study still has data blanks to a certain extent although dramatic improvements have been made. The objective of our AOD fusion framework, however, is to ensure that the recovered AODs possess spatiotemporal continuity rather than just filling up. When incorporating more AOD products derived from other SSO satellite sensors with different observation modes or scanning time into the Nested-BME fusion framework, the spatiotemporal coverage could be further improved with more recovered details about temporally continuous variation of aerosol, which requires future in-depth study.

Acknowledgment

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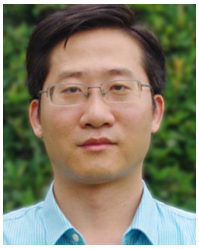
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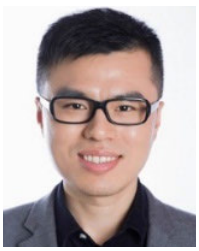
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